AQ8.5: Activity Questions 5 - Not Graded

×

This assignment will not be graded and is only for practice.
Level 1

1 point
Choose the correct options.

☐ Let $T: R^2 \to R^2$ be a linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

☐ Let $T: R^2 \to R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.

☐ Let $T: R^2 \to R^2$ be a linear transformation, where R^2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

☐ Let $T: R^2 \to R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let $T: R^2 \to R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
Let $T: R^2 \to R^2$ be an orthogonal linear transformation, where R^2 is the inner product space with respect to the inner product given by $< a, b > = 2a_1 b_1 + 5a_2 b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

1 point
If A is the matrix representation of an orthogonal transformation $T: R^n \to R^n$, then choose the set of correct statements.

☐ $AA^T = I$

☐ $A = A^T$

☐ AA^T is an upper triangular matrix

☐ AA^T is a lower triangular matrix

☐ $A^T = A^{-1}$

☐ $A^T = -A^{-1}$

☐ A^{-1} does not exist

☐ A^{-1} is also an orthogonal matrix

No, the answer is incorrect.
Score: 0

Accepted Answers:
$AA^T = I$
AA^T is an upper triangular matrix
AA^T is a lower triangular matrix
$A^T = A^{-1}$
A^{-1} is also an orthogonal matrix

1 point

If A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$, then choose the set of correct statements.

- [ ] The column vectors of A are orthogonal but not orthonormal.
- [ ] The column vectors of A are orthonormal.
- [ ] The row vectors of A are orthogonal but not orthonormal.
- [ ] The row vectors of A are orthonormal.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The column vectors of A are orthonormal.
The row vectors of A are orthonormal.

Level 2

1 point

Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ in R^3. Find a third vector in R^3 which is orthogonal to both the vectors in B.

- ( ) $(1, 1, -2)$
- ( ) $(1, -1, 2)$
- ( ) $(-1, 1, -2)$
- ( ) $(-1, -1, -2)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1, 1, -2)$

1 point

Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ and vector obtained in question 4 which is orthogonal to the set B. Form the columns of an orthogonal matrix U by scaling these vectors appropriately. Choose the correct option.

- ( ) $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$
- ( )

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$U = [3

√

1 3

√

1 3

√

1 2

√

−1 2

√

10 6

√

1 6

√

1 6

√

−2]

○

$U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$U = [111−110−11−2]

○

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$U = [3

√

1 3

√

1 3

√

1 2

√

−1 2

√

10 6

√

−1 6

√

1 6

√

−2]

No, the answer is incorrect.
Score: 0
Accepted Answers:

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$U = [3

√
1 3
√
1 3
√
1 2
√
−1 2
√
10 6
√
1 6
√
1 6
√

−2]

1 point

Consider a vector $v = (3, 0, 4)$v = (3, 0, 4) and UU from Question 5. Choose the correct option using the dot product as the standard inner product.

- $\| v \| = \| Uv \|$ ‖v‖ = ‖Uv‖
- $\| v \| \neq \| Uv \|$ ‖v‖ = ‖Uv‖
- $2\| v \| = \| Uv \|$ 2‖v‖ = ‖Uv‖
- $\| v \| = 2\| Uv \|$ ‖v‖ = 2‖Uv‖

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\| v \| = \| Uv \|$ ‖v‖ = ‖Uv‖

1 point

Let the matrix representation of a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$.

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$. Choose the set of correct statements.

- [ ] T is an orthogonal transformation.
- [ ] T is not an orthogonal transformation.
- [ ] B is a real symmetric matrix.
- [ ] $U^{-1}BU$ is a diagonal matrix.
- [ ] $U^{-1}BU$ is a scalar matrix.
- [ ] $U^{-1}BU$ is an upper triangular matrix.
- [ ] $U^{-1}BU$ is a lower triangular matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T is an orthogonal transformation.
B is a real symmetric matrix.
$U^{-1}BU$ is a diagonal matrix.